

KUSHAGRA SINGH KUSHWAH

+91 8109703612 | Nagpur, Maharashtra | kushagrasinghkushwah46@gmail.com
[linkedin.com/in/kushagrakushwah](https://www.linkedin.com/in/kushagrakushwah) | github.com/kushagrakushwah | kushagrask-portfolio.vercel.app

EDUCATION

Visvesvaraya National Institute of Technology, Nagpur Bachelor of Technology in Computer Science and Engineering	2024 – 2028
Kendriya Vidyalaya, Embassy of India, Kathmandu, Nepal Class XII	2024

EXPERIENCE

IITB Trust Lab, Indian Institute of Technology Bombay <i>Security Engineering Intern</i>	Mumbai, Maharashtra May 2026 – July 2026
- Engineered SecureForge, a containerized Breach & Attack Simulation platform using FastAPI, Next.js 14, and Kubernetes, orchestrating modular attack simulations with parallel execution and isolated WebSocket event streams per run. - Implemented MITRE ATT&CK ID mapping per finding and SOC coverage scoring; integrated ELK Stack telemetry with Nmap-based subnet discovery across configurable port ranges and timing profiles.	
IIT Bombay – Prof. Uday Khedker (Dept. of CSE) <i>Undergraduate Researcher – Points-To Analysis & Data Flow Frameworks</i>	Remote / VNIT Nagpur March 2026 – Present
- Collaborating with Prof. Uday Khedker (IIT Bombay) and Prof. Swati Jaiswal (VNIT) on unpublished points-to analysis research; implementing novel PTA variants on a private codebase targeting conference publication. - Validated all four analyses (Andersen's, Steensgaard's, FS-PTA, LFCPA) on PTA-Viz; studying bit-vector data flow frameworks to extend the visualizer with new analyses.	
Aarush GOC <i>Web Developer Intern</i>	Nagpur, Maharashtra May 2025 – June 2025
Engineered an AI chatbot using Python Deep Learning and Laravel MVC, automating 500+ daily queries and cutting resolution time by 40%; architected a PHP/MySQL employee portal eliminating 20+ hours of manual entry weekly at 99.9% uptime.	

PROJECTS

EnvAudit – Corporate SaaS Red-Teaming Environment for LLM Agents
- Built a complete RL training and evaluation system for a SaaS auditing agent using Qwen2.5-7B + GRPO in a partially observable enterprise environment with 3 difficulty levels and adversarial traps (CI/CD bot cancellation, annual contract penalty fees). - Engineered a FastAPI environment server with 7 tool-call APIs; designed dual reward functions (JSON format + brevity) collapsing avg. completion length from 128 tokens to ~10 tokens within 50 GRPO steps. - Trained via SFT on 600 oracle trajectories + GRPO reward shaping; GRPO model avoided the CI/CD bot trap on Task 3 (baseline score 0.01 to +0.020), demonstrating a learned safety property against irreversible actions. Tech: Python, FastAPI, Docker, Qwen2.5-7B, GRPO, RL, Pydantic, React/Vite, HuggingFace GitHub: github.com/kushagrakushwah/saas-openenv-audit
SecureForge – Breach & Attack Simulation Platform
- Built a self-hosted BAS platform using FastAPI and Next.js 14, orchestrating modular attack simulations with real-time WebSocket event streaming and MITRE ATT&CK coverage mapping per finding. - Deployed on Kubernetes with ELK Stack telemetry, HPA autoscaling, and Nmap-based subnet discovery across configurable port ranges and scan profiles. Tech: Python, FastAPI, Next.js, TypeScript, Docker, Kubernetes, ELK Stack, WebSockets, Nmap GitHub: github.com/kushagrakushwah/bas_v3
Route Optimisation & Demand Prediction System
- Engineered an MDVRP solver using Python and Google OR-Tools (CP-SAT) across 50+ geographical nodes; integrated OpenStreetMap APIs reducing theoretical fleet distance by 35%. - Conducted ML demand forecasting using Pandas and Scikit-Learn, boosting route prediction accuracy by 15% against standard algorithms. Tech: Python, Google OR-Tools, Scikit-Learn, Reinforcement Learning, RESTful APIs, Pandas GitHub: github.com/kushagrakushwah/Route-Optimisation-OR-Tools

TECHNICAL SKILLS

Languages	C, C++, Python, JavaScript, TypeScript
Web & APIs	HTML, CSS, PHP, Laravel MVC, FastAPI, Next.js, REST APIs, WebSockets
Security & AI	MITRE ATT&CK, BAS/Red-Teaming, LLMs/AI Agents, GRPO/RL, Computer Vision (OpenCV)
Data & ML	Pandas, NumPy, Scikit-learn, TensorFlow, HuggingFace, MediaPipe
DevOps	Docker, Kubernetes, ELK Stack, Git, GitHub, MySQL
Coursework	DSA, OS, DBMS, OOPs (C++), AI, Discrete Mathematics, Probability & Statistics

POSITIONS OF RESPONSIBILITY

Training and Placement (TnP) Coordinator – CSE Branch, VNIT Nagpur	2026 – Present
Vice President & Core Member – SHELL (Cyber Security Club), VNIT Nagpur	2025 – Present
- Promoted to Vice President; mentoring a 16-week blockchain research project (BlockVerify – Fake Product Identification using Solidity, Truffle, Web3.js) with junior students, targeting IEEE conference publication. - Spearheaded a Maharashtra-level CTF, architecting challenges in Cryptography and Web Exploitation; conducted a Wire-shark workshop on packet sniffing, network protocol analysis, and traffic telemetry.	
Maintenance Councillor – First-Year Hostel, VNIT Nagpur	2024 – 2025

HONORS & ACHIEVEMENTS

Grand Finalist, Meta PyTorch OpenEnv Hackathon x Scaler: Selected from 52,000+ developers to present <i>EnvAudit</i> at the Bangalore finale, competing for direct interviews with Meta & Hugging Face AI teams. Summer of Code, IIT Bombay Trust Lab: Awarded a stipended research internship (Ref. ACR-TR-26-122) for building the SecureForge BAS platform, selected via application and technical interview.	
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